Hayley Williams

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Employment

Beus Prize Postdoctoral Research Fellow — Sept 2025 - Present
Beus Center for Cosmic Foundations
Arizona State University – School of Earth and Space Exploration

Education

Ph.D. University of Minnesota — Sept 2019 - June 2025
Astrophysics
Thesis title: *The high-redshift Universe as seen through galaxy-cluster gravitational lenses*

B.A. University of California, Berkeley — Aug 2015 - May 2019
Astrophysics and Spanish Linguistics

First-Author Publications

A Magnified Compact Galaxy at Redshift 9.51 with Strong Nebular Emission Lines — April 2023
Hayley Williams, Patrick Kelly, Wenlei Chen, et al. Science, Volume 380, Issue 6643, pp. 416-420
10.1126/science.adf5307

Sp1149. I. Constraints on the Balmer $L - \sigma$ Relation for H II Regions in a Spiral Galaxy at Redshift 1.49 Strongly Lensed by the MACS J1149 Cluster — July 2024
Hayley Williams, Patrick Kelly, Wenlei Chen, et al. The Astrophysical Journal, Volume 969, Issue 1, id.54
10.3847/1538-4357/ad4464

Sp1149. II. Spectroscopy of HII Regions near the Critical Curve of MACS J1149 and Cluster Lens Models — July 2024
Hayley Williams, Patrick Kelly, Wenlei Chen, et al. The Astrophysical Journal, Volume 967, Issue 2, id.92
10.3847/1538-4357/ad4354

JWST's PEARLS: Temperatures of Nine Highly Magnified Stars in a Galaxy at Redshift 0.94 and Simulated Stellar Population Dependence on Stellar Metallicity and the Initial Mass Function — January 2026
Hayley Williams, Patrick Kelly, et al. The Astrophysical Journal, Volume 996, Issue 1, id.105
10.3847/1538-4357/ae1966

JWST's PEARLS: A Candidate Massive Binary Star System in a Lensed Galaxy at Redshift 0.94 — February 2026
Hayley Williams, Patrick Kelly, et al. The Astrophysical Journal, Volume 997, Issue 2, id.292
10.3847/1538-4357/ae2003

Accepted Observing Proposals

JWST **COHORTS: COsmic HORseshoe Transient Survey** — March 2026, Cycle 5
GO 10417, PI Sung Kei Li, *Co-I Hayley Williams*

HST **Stellar Initial Mass Function and Dark Matter from [OII] and Paschen-alpha Emission Maps Of the Giant Dragon Arc in the Hubble Frontier Fields Galaxy Cluster Abell 370** — July 2025, Cycle 33
GO 18106, PI Patrick Kelly, *Co-I Hayley Williams*

Keck **Interpretation of HST and JWST Imaging of Individual Magnified Stars at z=0.725 using OSIRIS AO Near-Infrared Spectroscopy** — June 2025, 2025B
2025B U232 , PI Alex Filippenko, *Co-I Hayley Williams*

JWST **The Dragon survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle Multi-cadence Microlensing at z=0.73** — March 2025, Cycle 4
GO 7345, PI Yoshinobu Fudamoto, *Co-I Hayley Williams*

JWST **JWST as a time machine: weighting the carbon produced exclusively by massive stars** — Feb 2024, Cycle 3
GO 6073, PI Annalisa Citro, *Co-I Hayley Williams*

JWST **Early stars – Properties of lensed stars at z $\sim$ 7** — Feb 2024, Cycle 3
GO 5058, PI Lucas Furtak, *Co-I Hayley Williams*

Keck **Flashlights: Many Extremely Magnified Individual Stars as Probes of Dark Matter and Stellar Populations to Redshift z = 2** — June 2022, 2022B
PID 75/2022B N181, PI Patrick Kelly, *Co-I Hayley Williams*

JWST **Imaging and Spectroscopy of Three Highly Magnified Images of a Supernova at z=1.5** — Oct 2022, Cycle 1
DD 2767, PI Patrick Kelly, *Co-I Hayley Williams*

JWST **Imaging and Spectroscopic Follow-up of a Supernova at Redshift z=3.47** — Oct 2022, Cycle 1
DD 2756, PI Wenlei Chen, *Co-PI Hayley Williams*

Selected Talks

Highly magnified individual stars at cosmological distances: nine massive stars detected as transient events in the Warhol galaxy at $z = 0.94$ — Sept 2025, Tucson, AZ
Invited colloquium presentation at Steward Observatory, University of Arizona

Highly magnified stars in the Warhol arc — April 2025, Cambridge, MA
Invited talk at "Strong Lensing in the Next Decade"
Center for Astrophysics, Harvard & Smithsonian

Constraints on the Balmer $L - \sigma$ relation for H II regions in a lensed spiral galaxy at $z = 1.49$ — June 2024, Madison, WI
Contributed Talk at the 244th meeting of the American Astronomical Society

An extremely compact yet highly star-forming magnified galaxy at $z = 9.51$ — May 2024, Reykjavik, Iceland
Contributed talk at "Extreme galaxies in their extreme environments at extremely early epochs"

A highly magnified and extremely compact galaxy at $z = 9.51$ with strong emission lines — July 2023, Sesto, Italy
Contributed talk at "The James Webb Space Telescope turns one: the birth and growth of galaxies"
Sexten Center for Astrophysics

Awards

Beus Prize Postdoctoral Fellowship — Sept 2025
Beus Center for Cosmic Foundations
Arizona State University – School of Earth and Space Exploration

Robert O. Pepin Fellowship — May 2024
University of Minnesota – School of Physics and Astronomy

Outreach, Leadership, and Service

Our Expanding Universe — June 2025, Minneapolis, MN
Bell Museum Planetarium Show
Assisted with script writing and show production

Interpreting the Invisible Universe — August 2024, Minneapolis, MN
A Project Space Grant Art Exhibition
Science consultant for artist Holly Streekstra

Universe in the Park — August 2024, Minneapolis, MN
Minnesota Institute for Astrophysics Public Outreach Event
Gave a public astronomy presentation and led a public observing night

Beus Seminar Committee — Sept 2025 - Present
Arizona State University

Journal Club Organizer — Sept 2023 - May 2025
University of Minnesota